

Received 29 October 2025, accepted 17 November 2025, date of publication 25 November 2025,
date of current version 4 December 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3636586

APPLIED RESEARCH

Housing Price Prediction Using AI Agent With Causal Inference for Societal Factor Selection

RAY-I CHANG¹, (Member, IEEE), DONG-MING HSIEH¹, AND TSUNG-SHENG LIN

Department of Engineering Science and Ocean Engineering, National Taiwan University, Taipei 10617, Taiwan

Corresponding author: Dong-Ming Hsieh (d08525001@ntu.edu.tw)

ABSTRACT Current housing price prediction (HPP) models face difficulties in identifying and explaining the causal impact of multidimensional societal factors. Existing statistical and deep-learning approaches rely heavily on correlation-based modeling, which limits their interpretability and adaptability to dynamic urban conditions. To address this gap, this study proposes an autonomous artificial intelligence (AI) agent framework that integrates a cyber-physical-social system (CPSS) ontology with a structural causal model (SCM) for intelligent societal factor selection and interpretable housing price prediction. AI agents operate through a self-adaptive cycle of perception, causal reasoning, and iterative learning. The CPSS ontology organizes complex societal variables into six core dimensions, enabling structured causal analysis, whereas SCM performs Bayesian-network-based factor screening. The selected factors are then used in an XGBoost model to enhance prediction accuracy and interpretability. The proposed CPSS-SCM framework provides quantitative evidence of the varying impacts of societal factors across urban environments, supporting transparent and region-specific real estate market analyses. Experiments conducted on 23,901 real estate transaction records from four districts in Taipei demonstrated significant improvements over traditional housing-factor-only models: the mean absolute percentage error (MAPE) decreased by 5%, the root mean squared error (RMSE) was reduced by 11%, and the coefficient of determination (R^2) score improved by 4%. The R^2 score consistently remained between 0.92 and 0.93 across districts, indicating excellent robustness. This integration represents a novel causal-predictive pipeline that improves both model transparency and predictive robustness in urban analytics.

INDEX TERMS AI agent, causal inference, structural causal model, CPSS, housing price prediction, societal factor, factor selection, Bayesian networks, XGBoost.

I. INTRODUCTION

Housing prices have long been a critical socioeconomic indicator, as real estate markets respond sensitively to multifaceted influences including economic cycles, social dynamics, and policy interventions. Social dynamics, geographical location, transportation and living amenities, and external variables such as government policies and macroeconomic indicators all affect housing price trends [1], [2]. The complex interactions among these heterogeneous factors make HPP highly challenging [3]. Recent studies have shown that incorporating macroeconomic variables can improve model accuracy [4], such as the use of power-law models to

analyze the housing bubble phenomenon in Las Vegas, United States [5]. However, traditional approaches often encounter efficiency limitations and reduced precision when handling large-scale, multidimensional data.

With the rapid advancement of AI, particularly the emergence of AI agents, new approaches and research perspectives have been introduced into HPP. Machine learning (ML) has become a mainstream tool for housing market analysis, owing to its advantages in large-scale data processing. Nevertheless, most ML models suffer from limited model transparency, which restricts interpretability and thereby reduces their usefulness in decision making.

The integration of CPSS and SCM is nontrivial because these two frameworks complement distinct analytical dimensions. The CPSS ontology provides a structured

The associate editor coordinating the review of this manuscript and approving it for publication was Kit Yan Chan¹.

representation for organizing social and environmental complexities, whereas the SCM focuses on extracting directional causal relationships from heterogeneous variables. This combination bridges the gap between descriptive societal knowledge and causal inference, thereby enabling the transformation of unstructured social data into interpretable causal structures.

The proposed AI agent exhibits autonomous intelligence through adaptive causal reasoning and self-updating knowledge. By continuously learning from the prediction residuals, the agent dynamically refines the causal weights and feature selections to accommodate the shifting societal trends, policies, and economic cycles.

The main contributions of this study are as follows:

- 1) Novel AI agent framework integrating CPSS ontology and SCM for interpretable causal predictive modeling.
- 2) Region-specific validation across the four Taipei districts demonstrated a consistent performance and explainability.
- 3) A computationally efficient implementation balances interpretability, accuracy, and scalability for real world HPP applications.

Overall, the proposed framework offers a systematic, operational, and practically valuable tool for HPP, representing one of the first integrated applications of causal inference and AI agents in real estate analysis. The remainder of this paper is organized as follows: Section II reviews the related literature, Section III introduces the research framework, Section IV presents the experimental results, and Section V concludes with future directions.

II. RESEARCH FRAMEWORK

A. AI AGENT AND CPSS

AI agents are regarded as advanced application tools that interact with large language models (LLMs). They operate in open environments and integrate multimodal inputs to achieve higher-level perception capabilities [6]. These attributes empower AI agents not only to handle basic data-cleaning tasks but also to perform complex computations and analyses, serving as the backbone of autonomous task execution [7], [8], [9]. In this study, AI agents were applied to societal factor analysis and causal inference. Through sustained learning and refinement, they adapt to rapid changes in societal factors, thereby enhancing the model's data-analysis capabilities and interpretability.

The CPSS extends the concept of cyber-physical systems (CPS) by incorporating social phenomena [10], [11], [12]. CPSS have facilitated advancements in various applications, including traffic management for autonomous vehicles [13], recommendation systems for optimizing travel routes [14], and behavioral analysis models for predicting future actions [15]. This study investigated the societal factors that influence housing prices. As illustrated in Fig. 1, CPSS ontology [16] was adopted to classify societal factors into six categories: Devices, Services, Activities, Persons, Roles, and

Events. This classification framework supports a systematic analysis of the relationship between societal factors and housing prices.

In this study, the CPSS ontology is defined as a knowledge-based hierarchical model representing the dynamic interactions among cyber, physical, and social components within an urban system. Ontology construction followed a three-phase process.

- 1) Conceptual design: The six dimensions—Device, Service, Activity, Person, Role, and Event—were defined following Smirnov et al. to establish the CPSS ontology foundation.
- 2) Data mapping: Heterogeneous data sources (e.g., government open data, Google Maps APIs, and economic indicators from the Central Bank of Taiwan) were semantically aligned with their corresponding ontology nodes to ensure data consistency and interoperability.
- 3) Interaction modeling: Interdependencies among categories were captured using a hierarchical relation matrix to formalize causal links (e.g., Role $\rightarrow$ Activity $\rightarrow$ Event), which frequently influence housing market dynamics.

This hierarchical and relational design allows the CPSS ontology to serve as a semantic bridge between raw data and causal structure modeling, supporting both factor interpretability and consistency during causal inference.

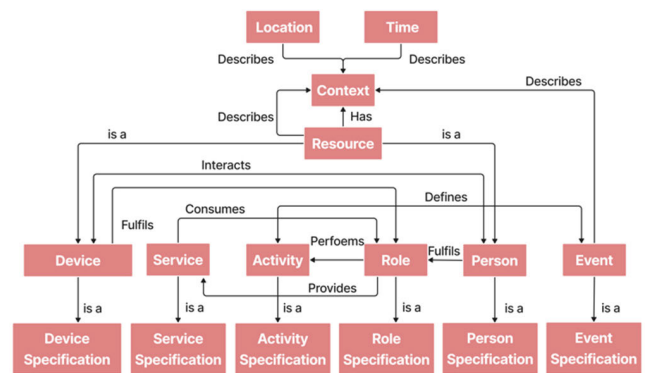


FIGURE 1. The CPSS ontology for classifying societal factors into six categories: Device, Service, Activity, Person, Role, and Event.

B. HPP AND SOCIETAL FACTORS

HPP has long been a critical research topic in economics and information science. Recent studies have shown that refining prediction models for specific administrative districts can significantly enhance accuracy [17], whereas incorporating macroeconomic indicators further improves the predictive performance [18]. Regarding political variables, previous research has included the ruling party of a city to examine the potential influence of central and local government parties on housing price trends. These studies indicate that property characteristics generally exert greater influence on housing prices than political factors.

Societal factors, encompassing events and phenomena arising from social activities, directly or indirectly shape individual behavior and decision making, thereby significantly influencing the real estate market. Prior research has identified several societal factors closely associated with housing prices, including the following.

- 1) Macroeconomic indicators such as the consumer price index (CPI) [19].
- 2) Corporate relocation events, such as TSMC's establishment in Tainan, positively affect housing prices in surrounding areas [20].
- 3) Vacant home tax policies, which most significantly impact mid-rise apartments, lead to the highest price increases [21].
- 4) Transportation diversity, where a mix of options enhances population mobility and convenience, elevates housing prices near transit hubs [22].
- 5) Pandemic severity, with the COVID-19 pandemic suppressing real estate transactions [23].
- 6) Spatial correlations exist because housing prices across counties exhibit dependencies, with factors such as dependency ratios and household disposable income showing positive effects [24].

C. CAUSAL INFERENCE AND BAYESIAN METHODS

Causal inference is the process of determining the independent effect (outcome) of a specified phenomenon (cause) within a system. Unlike correlation analysis, causal inference examines how a response variable reacts to changes in a causal variable, with the primary goal of establishing causal relationships between variables and leveraging these relationships for prediction and inference.

Maximum likelihood estimation (MLE) and Bayesian estimation are two commonly used approaches for parameter learning in causal inference. Because MLE is prone to overfitting, which reduces model generalization on new data, Bayesian estimation is often preferred in practice owing to its robustness in small-sample or missing-data scenarios. To enhance the structural search efficiency, this study employs a hill climbing algorithm, which accelerates the computation while maintaining the accuracy [25]. Additionally, the Bayesian-Dirichlet (BD) score serves as a structural evaluation metric suitable for datasets with prior distributions or limited scales [26]. Bayesian estimation is particularly effective for constructing Bayesian network models.

In causal inference, methods are applied when one variable potentially influences another. The causal Markov condition (CMC) stipulates that a variable X_i is conditionally independent of non-descendant variables, with its behavior described by its parents X_j and an independent error term ϵ_j , as shown in Equation (1).

$$X_j \prod X_i | Pa(X_i), \epsilon_j \quad (1)$$

Extending this, the structural equation model (SEM) uses a set of equations to depict causal relationships among multiple

variables, each corresponding to a variable's CMC, as shown in Equation (2).

$$P(X_1, \dots, X_j) = \prod_{j=1}^J P(X_j | Pa(X_j), \epsilon_j) \quad (2)$$

The joint probability distribution is the product of the conditional distributions of all variables [27]. Integrating historical causal insights with SEM has led to the development of SCM, visualized via directed acyclic graphs (DAGs), to represent complex variable relationships. In SCM, the variable Y is determined by other variables and an error term through a specific functional relationship, where the error term captures exogenous stochastic factors, as shown in Fig. 2. The value of each variable is derived from its parents and error term, as expressed in Equation (3).

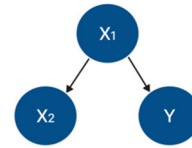


FIGURE 2. DAG for SCM.

$$\begin{aligned} Y &= f_Y(X_1, X_2, \epsilon_Y) \\ X_1 &= f_{X_1}(\epsilon_{X_1}), X_2 = f_{X_2}(X_1, \epsilon_{X_2}) \end{aligned} \quad (3)$$

1) INTERVENTION AND CONFOUNDING VARIABLES

Intervention involves deliberately altering a variable's value in an SCM, denoted by the *do*-operator (e.g., $\text{do}(X_2 = t)$), which sets variable X_2 to a specific value t while disregarding its original causal dependencies [27]. The intervention distribution describes the resulting probability distribution of the other post-intervention variables. As illustrated in Fig. 3(a), removing the direct effect of a variable on Y leaves the distribution of Y determined solely by the remaining paths, reflecting the causal effect of fixing X_1 , as illustrated in Fig. 3(b). This approach is particularly suitable for estimating direct causal effects.

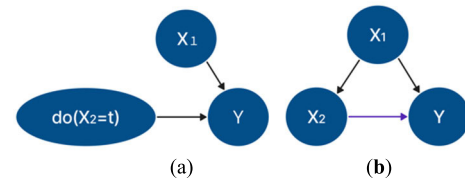


FIGURE 3. (a) Causal diagrams describing interventions. (b) Causal diagrams with confounding variable X_1 .

Confounding bias arises when an unobserved variable simultaneously influences both the cause and outcome, thereby distorting the true relationship. These variables are termed confounders. In a causal graph, a confounder links cause and outcome. Without controlling for confounders, the

observed conditional distributions may contain bias, whereas the intervention distributions may isolate true effects. The average treatment effect (ATE) quantifies the expected change in Y due to an intervention, as defined in Equation (4).

$$\begin{aligned} \text{ATE} &= E[Y|do(X_2 = 1)] - E[Y|do(X_2 = 0)] \\ \text{ATE} &= P(B|do(A = 1)) - P(B|do(A = 0)) \\ &= \sum_T (P(B|do(A = 1), T = t) \\ &\quad - P(B|do(A = 0), T = t))P(T = t) \\ &= \sum_T (P(B|A = 1, T = t) - P(B|A = 0, T = t)) \\ &\quad \times P(T = t) \end{aligned} \quad (4)$$

2) CAUSAL IDENTIFICATION AND BACKDOOR CRITERION

Causal inference aims to derive causal quantities (intervention effects) from statistical quantities (correlations or conditional distributions) through a process known as causal identification. The backdoor criterion provides a method to eliminate confounding bias by addressing non-causal paths (backdoor paths) from cause to effect, as illustrated in Fig. 3(b). If variable X_1 affects both X_2 and Y (via $X_1 \rightarrow X_2$ and $X_1 \rightarrow Y$) and there is a direct path $X_2 \rightarrow Y$, a backdoor path exists. By controlling for such confounders, this study estimated the true causal effects from observational data, thereby distinguishing causality from correlation.

D. XGBOOST

XGBoost is an advanced implementation of the gradient boosting decision-tree framework [28]. Designed to minimize prediction bias through careful optimization of the objective function, XGBoost employs classification and regression trees (CART) as its base learners. It supports a wide range of objective functions, including regression, classification, and ranking. To enhance generalization and mitigate overfitting, XGBoost incorporates L1 and L2 regularization techniques, along with column and row sampling strategies, to increase data diversity and reduce overfitting risks. Furthermore, it achieves substantial computational efficiency through compressed tree structures and parallel computing, enabling rapid training even on large datasets.

Owing to its high accuracy and computational speed, XGBoost has demonstrated exceptional performance across various domains and in data-science competitions. For example, in real estate market analysis, it exhibits superior predictive capabilities [29]; it outperforms traditional decision trees and random-walk models in corporate credit risk assessment [30], and has been successfully applied to wafer yield prediction [31]. These examples highlight the robustness, flexibility, and broad applicability of XGBoost.

III. CONSTRUCTION

A. AI AGENT DESIGN

In this study, an AI agent was designed to improve both the accuracy and interpretability of HPP by performing

automated data processing, causal analysis, and predictive modeling. Its main objectives include ensuring data quality, selecting key societal factors, optimizing models, and automating operational processes. The architecture integrates (1) a preprocessing module (e.g., data cleaning and CPSS ontology construction), (2) causal inference tools (e.g., hill climbing algorithm with BD score, Bayesian estimation of CPTs, DoWhy, and causal-learn for SCM and backdoor-criterion analysis), and (3) predictive modeling using XGBoost supported by a feedback mechanism for dynamic adjustments. Fig. 4 illustrates the workflow, demonstrating how preprocessing, causal inference, and prediction are integrated within the agent's planning and action cycle.

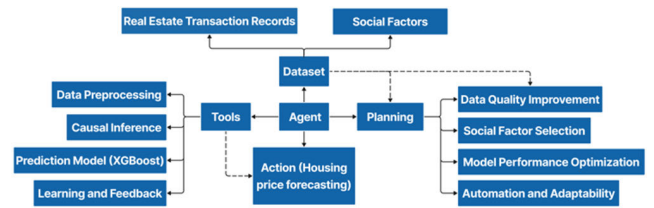


FIGURE 4. Workflow of the proposed AI agent.

In this study, the AI agent is designed as an autonomous learning system that executes an adaptive intelligence cycle composed of four stages:

- 1) Perception: Collects and preprocesses transactional and societal data via the CPSS ontology to ensure data integrity and factor consistency.
- 2) Reasoning: Learns directed causal graphs and Bayesian parameters using the hill climbing algorithm and BD scoring to infer factor dependencies.
- 3) Action: Generates predictive models using XGBoost and evaluates them using the MAPE, RMSE, and R^2 metrics.
- 4) Learning: Compares prediction errors with previous results and dynamically updates the causal weights and factor sets through a feedback mechanism.

This cyclical learning loop allows the AI agent to adjust autonomously to new data distributions or external shocks (e.g., policy changes), ensuring continual improvement in both accuracy and interpretability. The adaptive learning process of the AI agent is described in Algorithm 1, which details the iterative causal-predictive feedback mechanism.

B. SOCIETAL FACTOR CLASSIFICATION AND DATA DISCRETIZATION

1) SOCIETAL FACTOR CLASSIFICATION

This study adopted the CPSS ontology [16] to categorize the selected factors, and the results are shown in Fig. 5. The specifications for each category are summarized as follows:

- a) Device: Includes public transportation facilities that enhance convenience, such as MRT stations, bus stations, and YouBike stations. These facilities improve urban mobility and influence housing prices either directly or

Algorithm 1 Adaptive AI Agent Learning Process.

Step	Description
Input	Dataset D , societal factors S
Output	Updated model M^* and factor weights W^*
1	Preprocess D and map S via CPSS ontology
2	Learn causal graph $G \leftarrow \text{hill climbing}(D, S)$ using BD score
3	Estimate Bayesian parameters for G
4	Train XGBoost model M on selected causal factors
5	Evaluate prediction error $*E=$
6	If $E > \varepsilon$ (error threshold): update $W \leftarrow \text{Adjust}(\text{factor_importance}, \text{ATE})$; re-learn G ; retrain M
7	Return optimized model M^* and updated weights W^*

indirectly by enhancing accessibility. The number of such facilities often serves as a key indicator of regional convenience.

- b) Service: Comprises essential lifestyle services, including food, clothing, housing, education, and entertainment, such as educational resources, healthcare, public safety, supermarkets, and cultural or recreational facilities. These services enhance quality of life and convenience, significantly impacting housing market prices.
- c) Activity: Consists of macroeconomic indicators reflecting overall economic health and performance, including the central bank interest rate, consumer price index, household disposable income, engineering construction index, money supply, loan rate, stock index, economic growth rate, and unemployment rate. These indicators shape the economic environment, thereby affecting the housing market supply, demand, and pricing.
- d) Person: Includes demographic and structural indicators such as the elderly dependency ratio and total population, which describe regional population dynamics and their influence on housing demand and prices.
- e) Role: Represents entities that influence housing prices through policy-making, economic activities, investment, and development, such as the Taipei City Government, central government, large corporations, and construction firms. These roles shape the supply-demand dynamics and market expectations.
- f) Event: Includes events affecting economic activities, social behavior, and market expectations, such as daily COVID-19 case counts. Such events can trigger market fluctuations, alter demand, and impact housing price.

2) DATA DISCRETIZATION

In Bayesian network models, parameters must be represented as discrete values, requiring discretization of continuous variables prior to training. Discretization converts quantitative data into qualitative forms, thereby improving ML performance and interpretability [32]. For large datasets with complex factor relationships, discretization reduces variable cardinality, accelerates iterations, enhances model robustness, mitigates the impact of outliers and extreme values, and introduces nonlinearity to improve expressive power [33]. Common discretization methods include the uniform-count

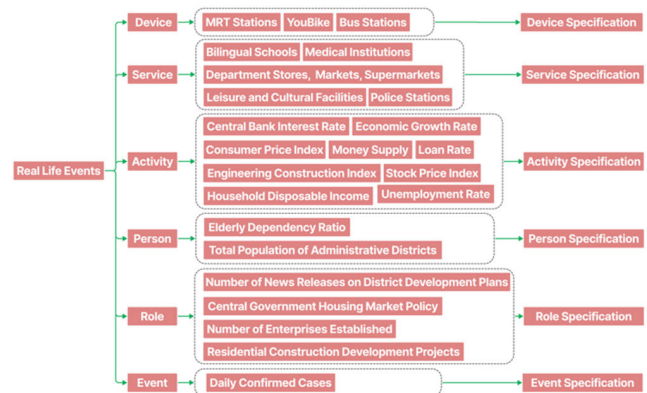


FIGURE 5. The CPSS ontology.

method (equal frequency) and the uniform-width method (equal width). This study employs the uniform-width method to preserve the original data distribution.

The uniform-width method divides data into equally spaced intervals by specifying the number of bins and calculating the separation points. For n bins, the separation points were computed using (5).

$$w = \frac{max - min}{n} \quad (5)$$

where max and min denote maximum and minimum values, respectively. This formula segments data into n equal intervals. According to the literature [34], dividing data into five to 20 intervals facilitates rapid preliminary analysis. In this study, 10 equal intervals were adopted to discretize the 23 selected variables.

3) APPLICATION SERVICES: CAUSAL INFERENCE AND BAYESIAN NETWORKS

This study leverages causal inference and Bayesian network theories to design a four-step sequential process, providing a clear and systematic framework for methodological explanations and potential applications in related fields. The research framework illustrated in Fig. 6 comprises four key components.

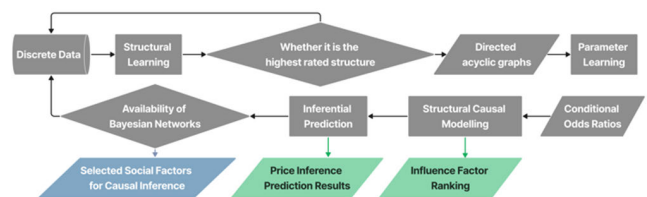


FIGURE 6. Causal inference and Bayesian networks with application processes.

- a) **Structural Learning of DAG:** The hill climbing algorithm is employed for structural learning, with the BD score serving as the evaluation metric. Starting from an empty graph, the algorithm assigned an initial score to the network. Two nodes are randomly selected to establish

a directed edge. If the new connection improves the score, then the updated network becomes the current best solution. Neighboring solutions are generated via three operations: (1) removing a directed edge, (2) adding a directed edge, or (3) reversing the direction of an existing edge. The option that yielded the greatest score improvement over the current best solution is adopted. This process is repeated until no further score enhancement is possible, resulting in DAG.

- b) **Parameter Learning for CPTs:** This phase estimates conditional probability tables (CPTs) based on the established network structure. Parameter learning calculates the conditional probability distribution of each variable relative to its parent nodes using the sample data. To mitigate overfitting commonly encountered with MLE, Bayesian estimation is applied, assuming the parameters follow a Dirichlet distribution and compute their maximum a posteriori estimates to generate CPTs. The importance of parameter learning in model construction and its impact on subsequent analyses were also discussed.
- c) **Validating Causal Relationships with SCM:** After structural and parameter learning, a probabilistic DAG is obtained. However, causal relationships between nodes remain unconfirmed, preventing direct causal inference or prediction. To establish causality, SCM [26] combined with the backdoor criterion was employed to infer causal effects from observational data. By controlling for variables along backdoor paths, causal identification was achieved, enabling the estimation of true causal effects and producing a weighted causal DAG.
- d) **Inference, Prediction, and Factor Importance Ranking:** The constructed Bayesian network is used for inference and prediction, while ranking the importance of the influencing factors. Inference algorithms, evaluation metrics, and the generation of predictive outcomes along with factor weight analysis are described. The practical implications of these results are discussed, providing a rigorous and reproducible framework for applying Bayesian networks and causal inference within the context of this research.

4) LEARNING AND IMPROVEMENT

After completing data cleaning and handling missing values, the AI agent enters the learning and improvement phase to refine its processing methods and enhance subsequent performance. Aligned with its objectives—constructing a high-quality dataset, improving HPP accuracy, and leveraging applied tools such as the CPSS ontology, Bayesian networks, causal inference, and XGBoost—this phase emphasizes self-adjustment and iterative optimization.

During data cleaning, the AI agent logs the outcomes of tasks such as missing-value imputation, duplicate removal, and outlier detection, and evaluates data quality based on metrics such as consistency and distributional properties. For example, if a method such as mean imputation distorts the data distribution, the agent adapts by switching

to context-based or model-predicted imputation techniques. Furthermore, after classifying societal factors using the CPSS ontology, the agent assesses the influence weights of the categories and adjusts the classification logic or incorporates additional variables via a feedback mechanism.

When applying causal inference and Bayesian networks, the AI agent evaluates the accuracy of the DAG and CPTs obtained from structural and parameter learning, using the backdoor criterion to validate causal effects. If confounding variables are inadequately controlled, the agent iteratively updates the model structure by introducing additional potential factors. Once the optimized data and causal relationships are fed into XGBoost, the AI agent analyzes prediction errors and factor-importance rankings to further learn the influence patterns of key factors.

Through repeated experimentation and feedback, the agent fine-tunes algorithm parameters, thereby achieving continuous improvement in predictive performance. This learning and improvement mechanism enables the AI agent to dynamically adapt to data characteristics and analytical requirements, delivering more accurate and interpretable HPP outcomes.

5) DATA SOURCES

The dataset employed in this study comprises two main components. The first component, provided by Leju Technology Co., Ltd., contains transaction-based data in the APR from four administrative districts of Taipei City—Nangang, Neihu, Beitou, and Wenshan—spanning July 2017 to June 2023, with a total of 23,901 records. Each record includes 23 fields that provide detailed information on housing transactions.

Table 1 presents a comparison between the societal factors adopted in this study and those employed in previous studies, highlighting the inclusion of several unique factors. Specifically, this study incorporates central government housing market policies and local administrative district development plans as political factors to capture their influence on housing prices. Conversely, it excludes factors that were found to exert less influence in previous research, such as the purchasing managers index, regular wages, and the several U.S.-based macroeconomic variables.

Money supply, U.S. economic growth rate, and the U.S. federal funds market rate have been shown to exert less impact on housing prices than property characteristics [35]. Moreover, because the consumer price index already accounts for housing components, it was not considered.

The second component consists of societal factor data collected from multiple official and public sources. Domestic macroeconomic indicators were obtained from the Statistical Information Network of Taiwan, the Central Bank of Taiwan, and the National Development Council. Central government housing policy data were provided by the Ministry of the Interior, whereas COVID-19 pandemic data were collected from the Taiwan Centers for Disease Control. Demographic information, including dependency ratios and district

TABLE 1. Compare the relevant societal factors in this study with those in other studies.

Societal factor	[4]	[23]	[24]	[35]	Ours
Loan Rate	O	O	O	X	O
Central Bank Interest Rate	X	X	X	X	O
Stock Price Index	X	X	O	X	O
Consumer Price Index	O	O	O	X	O
Consumer Price Rent Index	O	X	X	X	X
Engineering Construction Index	X	X	O	X	O
Money Supply	O	O	O	X	O
Economic Growth Rate	X	O	O	X	O
Unemployment Rate	O	X	O	X	O
Purchasing Managers Index	X	X	O	X	X
Regular Salary	X	X	O	X	X
Household Disposable Income	X	X	X	X	O
U.S. Money Supply	X	X	O	X	X
U.S. Economic Growth Rate	X	X	O	X	X
U.S. Federal Funds Market Rate	X	X	O	X	X
Daily Confirmed Cases	X	X	X	X	O
Total Population of Administrative Districts	X	X	X	X	O
Elderly Dependency Ratio	X	X	X	X	O
Central Government Ruling Party	X	X	O	X	X
Local Government Ruling Party of Northern Taipei and Taoyuan	X	X	O	X	X
Central Government Housing Market Policy	X	X	X	X	O
Number of News Articles on District Development Plans	X	X	X	X	O
Residential Construction Development Projects	X	X	X	X	O
Number of Enterprises Established	X	X	X	X	O
MRT Stations	X	X	X	O	O
Bus Stations	X	X	X	O	O
Youbike (Bike Sharing System)	X	X	X	X	O
Leisure and Cultural Facilities	X	X	X	O	O
Department Stores, Markets, Supermarkets	X	X	X	O	O
Police Stations	X	X	X	O	O
Medical Institutions	X	X	X	O	O
Bilingual Schools	X	X	X	X	O

population totals, was accessed from the Taipei City Data Platform. District development plan data were gathered from the Taipei City Government, Taipei City Industry Development Bureau, and Taipei City Urban Development Bureau. Transportation and daily life facility data were obtained from Google Maps. Educational resource data, including metrics such as the number of bilingual schools within each school district, were retrieved from the Taipei City Education Bureau and Leju Technology. A detailed summary of all items is presented in Table 2.

6) MODEL TRAINING

The model training process was divided into two components. The first involves training and prediction using the preprocessed dataset, whereas the second incorporates Bayesian networks to select societal factors prior to model training. The dataset was temporally partitioned, with the first 80% (July 2018–June 2022) used for training and the remaining 20% (July 2022–June 2023) reserved for testing.

Drawing on prior time-series ML studies for housing price prediction in Taiwan [24] and comparative analyses of

TABLE 2. List of data properties and items.

Data Field Name	Data Field Type
Administrative District	categorical
Subject of the transaction	categorical
Address	text
Community Name	text
Total area of land transferred (square meters)	float
Use Zoning or Designation	categorical
Date of Transaction	Int
Level of Transfer	Int
Total number of floors	Int
Building Type	categorical
Main Use	categorical
Completion Date	Int
Total area transferred (m ²)	float
Existing layout of the building-room	Int
Existing layout of the building-dining room	Int
Existing layout of building-bathroom	Int
Current layout of the building - cubicles	boolean
Total price (\$)	Int
Unit Price(\$/ping)	Int
Car park type	categorical
Total transferable area of car park (ping)	float
Total Car Park Price (\$)	Int
Public Ratio	float
Money Supply	float
Central Bank Interest Rate	float
Stock Price Index	float
Consumer Price Index	float
Structural Engineering Index	float
Loan Rate	float
Household Disposable Income	float
Economic Growth Rate	float
Unemployment Rate	float
Central Government Housing Market Policy	boolean
Daily Confirmed Cases	Int
Elderly Dependency Ratio	float
Total Population of Administrative Districts	Int
Number of News Articles on District Development plans	Int
Number of Enterprises Established	Int
Residential Construction Development Projects	Int
MRT Stations	Int
Bus Stations	Int
Youbike (Bike Sharing System)	Int
Leisure and Cultural Facilities	Int

gradient boosting models [26], this study fine-tuned the lambda and alpha parameters to determine the final hyperparameter configuration for XGBoost (Table 3). XGBoost is adopted for its computational efficiency and high predictive accuracy, whereas hyperparameter tuning ensures optimal model performance tailored to the characteristics of this dataset and research problem.

IV. RESULT

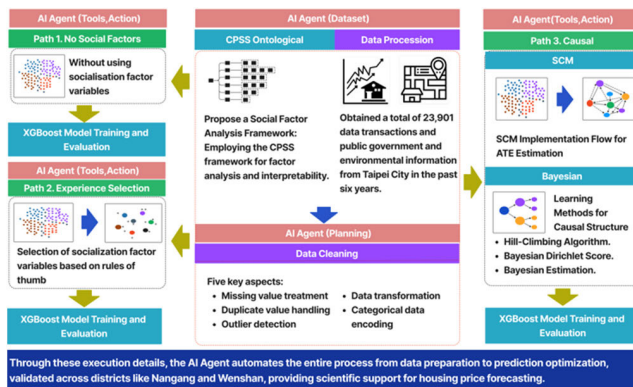
A. AI AGENT

This study developed a three-phase AI agent framework to enhance HPP: (1) cleaning 23,901 Taipei City transaction records (handling missing values, duplicates, and outliers), classifying societal factors (e.g., MRT stations, Central Bank interest rate) using the CPSS ontology, and discretizing 23 variables into 10 intervals; (2) generating weighted causal graphs; and (3) refining strategies based on prediction errors

TABLE 3. XGBoost hyperparameter settings.

Hyperparameter	Descriptions	Value
n_estimator	Number of boosting rounds (trees) used in the model.	100
learning_rate	Shrinks each tree's contribution to improve generalization.	0.2
max_depth	Maximum depth of each tree, controlling model complexity.	7
subsample	Fraction of training samples used for each tree.	0.95
Colsample_bytree	Fraction of features sampled when building each tree.	1
lambda	L2 regularization to reduce overfitting.	0.5
alpha	L1 regularization to enforce sparsity in leaf weights.	0.5

and factor weights to adapt to regional variations (e.g., Nangang vs. Wenshan). The proposed methodology not only improves forecasting accuracy through causal analysis but also provides insights into regional differences in the factors influencing housing prices, as illustrated in Fig. 7.


FIGURE 7. Explanation of AI agent workflow and module structure.

B. SCM AND INFERENCE PREDICTION RESULTS

Following structural and parameter learning of the Bayesian network, this study employs SCM to validate causal relationships, thereby ensuring the reliability of the derived results. SCM is implemented using the DoWhy and causal-learn packages, with the Backdoor Criterion applied to estimate the ATE from observational data. Multiple validation strategies available in DoWhy were adopted to enhance the robustness of causal inference, including the introduction of random confounders, placebo interventions, and subset validation, thereby rigorously assessing ATE stability.

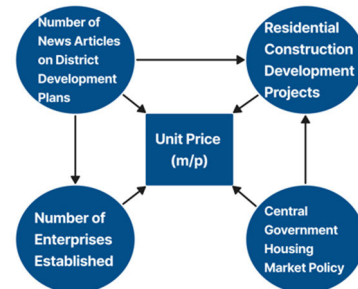
For example, increasing the number of MRT stations in Nangang increases housing prices by 13.176% (ATE = 0.13176). Similarly, the Central Government Housing Policy (ATE = 0.17445) and Loan Rates (ATE = 0.13832) exhibit significant effects in Nangang and Wenshan, respectively (Table 4). The weighted causal DAG further ranks societal factors, with Role-related factors (e.g., MRT stations)

dominating in Nangang, whereas activity-related factors (e.g., Loan Rates) dominate in Wenshan. These results enhance model interpretability and inform factor selection for subsequent XGBoost modeling.

TABLE 4. ATE of key societal factors.

Factor	District	ATE	P-value
MRT Stations	Nangang	0.13176	<0.01
Central Government Housing Market Policy	Nangang	0.17445	<0.01
Loan Rate	Wenshan	0.13832	<0.01
Number of Enterprises Established	Neihu	0.14089	<0.01
Leisure and Cultural Facilities	Beitou	0.15671	<0.01

Using Nangang as a case study, network structures based on CPSS societal factor classification are presented. Figs. 8 and 9 depict the causal network structures for the Role and Activity categories, respectively. The causal-learn package was used to construct a weighted causal DAG. Fig. 10 presents the weighted causal network structure for Nangang, integrating factor weights that influence the target variable and visually representing the effects of different societal factors on housing prices. Based on these weighted causal networks, the societal factors were ranked by importance, as summarized in Table 5. This ranking highlights not only the magnitude of each factor's effect on housing prices but also underscores the critical Role of societal influences in driving price variations.


FIGURE 8. Network structure of Nangang area role classification.

C. MODEL PREDICTION

1) INCREMENTAL FACTOR EVALUATION AND PERFORMANCE ANALYSIS

To validate the contribution of societal factors, a stepwise evaluation was performed by gradually increasing the number of selected factors ($k = 0-9$). The prediction results of the model based solely on transaction-level data in the actual price registration (APR) dataset served as the baseline, and societal factors were incrementally added to the XGBoost model. Fig. 11 shows the validation curve between k and MAPE, where the prediction error continuously decreases as more societal factors are included.

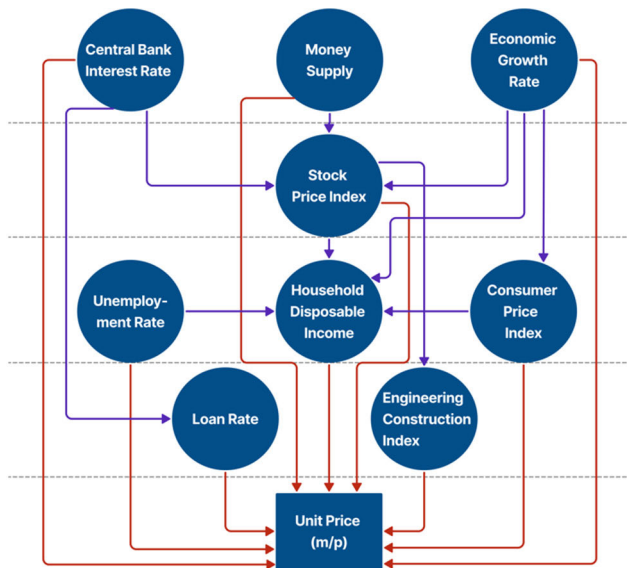


FIGURE 9. Network structure of Nangang area activity classification.

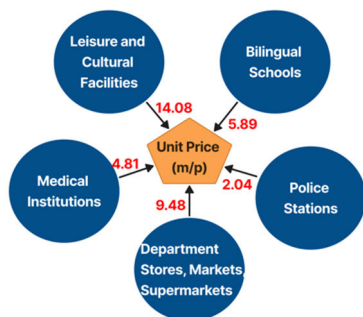


FIGURE 10. Directional No-circulation diagram of the causal relationships containing the weights of the target variables in the Nangang area.

The performance stabilizes at approximately $k = 7-9$, and according to the one-standard-error (one-SE) rule, $k = 9$ is identified as the optimal balance between accuracy and model complexity. The mean MAPE across the four districts decreased from 15.79% to 10.83%, representing a 31.4% improvement, whereas the R^2 score increased from 0.896 to 0.929 (Fig. 12). These results confirm the predictive stability and scalability of the proposed AI agent framework.

Unlike previous studies that relied on empirical rules and yielded inconsistent factor counts, the present analysis demonstrates that model performance peaks when nine societal factors are included, achieving optimal values across all evaluation metrics (MAPE, R^2 , and RMSE). Therefore, this study recommends selecting the top nine most influential societal factors as predictive variables, as summarized in Tables 6 and 7.

Further analysis revealed that the predictive improvement was more pronounced in Nangang District than in Wenshan District, suggesting that housing prices in emerging development zones are more sensitive to environmental changes and government investment activities. This empirical observation

TABLE 5. Importance ranking of societal factors derived from weighted causal DAG (Nangang & Wenshan).

	Nangang	Wenshan
1	Number of News Articles on District Development Plans	Loan Rate
2	Loan Rate	Money Supply
3	Residential Construction Development Projects	Residential Construction Development Projects
4	Central Government Housing Market Policy	Central Government Housing Market Policy
5	Money Supply	Consumer Price Index
6	Consumer Price Index	Number of News Articles on District Development Plans
7	Engineering Construction Index	Engineering Construction Index
8	MRT Stations	MRT Stations
9	Leisure and Cultural Facilities	Leisure and Cultural Facilities
10	Household Disposable Income	Department Stores, Markets, Supermarkets
11	Central Bank Interest Rate	Central Bank Interest Rate
12	Stock Price Index	Stock Price Index
13	Bus Stations	Bus Stations
14	Economic Growth Rate	YouBike
15	Department Stores, Markets, Supermarkets	Economic Growth Rate
16	Unemployment Rate	Household Disposable Income
17	YouBike	Unemployment Rate
18	Number of Enterprises Established	Bilingual Schools
19	Bilingual Schools	Medical Institutions
20	Medical Institutions	Police Stations
21	Total Population of Administrative Districts	Number of Enterprises Established
22	Elderly Dependency Ratio	Total Population of Administrative Districts
23	Police Stations	Elderly Dependency Ratio
24	Daily Confirmed Cases	Daily Confirmed Cases

aligns with the socioeconomic reality that newly developed areas (e.g., Nangang) exhibit greater responsiveness to societal dynamics than mature residential zones such as Wenshan. Additional validation conducted for Neihu and Beitou further confirmed the robustness of this finding.

TABLE 6. Forecast results: MAPE (%).

Number of Societal Factors	NANGANG	WENJSHEN	NEIHU	BEITOU
0	15.75%	15.83%	15.77%	15.79%
1	14.60%	14.97%	14.63%	14.64%
2	13.67%	14.19%	13.70%	13.71%
3	12.91%	13.49%	12.99%	12.95%
4	12.31%	12.88%	12.38%	12.34%
5	11.82%	12.34%	11.87%	11.85%
6	11.44%	11.88%	11.49%	11.48%
7	11.16%	11.50%	11.17%	11.19%
8	10.95%	11.20%	10.94%	10.97%
9	10.82%	10.93%	10.81%	10.83%

After presenting the causal-predictive results for each district, the consistency of the selected societal factors was analyzed to evaluate the generalizability of the proposed framework across different urban contexts. To quantify the degree of agreement between districts, pairwise overlaps and Jaccard similarities were computed for both the $k = 7$ and

TABLE 7. Forecast results: R^2 score.

Number of Societal Factors	NANGANG	WENJSHEN	NEIHU	BEITOU
0	0.8971	0.8879	0.8969	0.8967
1	0.9068	0.8943	0.9065	0.9064
2	0.9125	0.9000	0.9123	0.9122
3	0.9166	0.9051	0.9163	0.9164
4	0.9197	0.9096	0.9195	0.9196
5	0.9223	0.9135	0.9221	0.9222
6	0.9248	0.9167	0.9246	0.9247
7	0.9270	0.9192	0.9269	0.9268
8	0.9288	0.9211	0.9286	0.9287
9	0.9299	0.9224	0.9297	0.9298

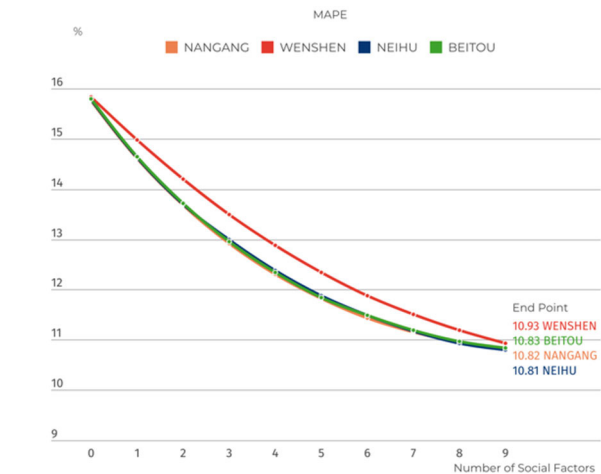


FIGURE 11. Trends in MAPE across different numbers of societal factors for four regions.

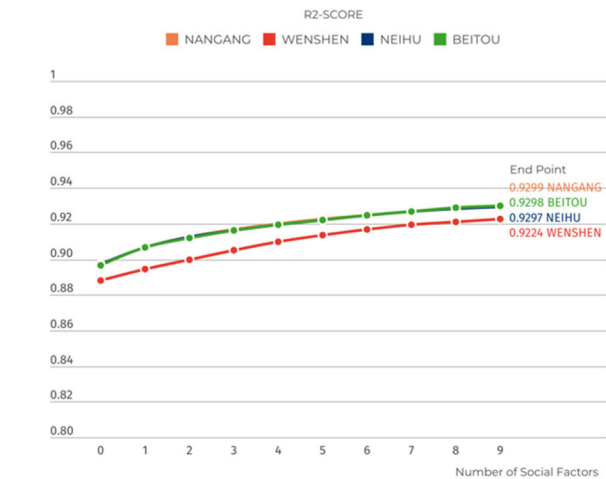


FIGURE 12. Trends in R^2 score across different numbers of societal factors for four regions.

$k = 9$ configurations, as summarized in Table 8. The results show an average Jaccard index of 0.80 for the top-9 configuration, indicating substantial cross-district stability in the selected factors.

Core determinants such as Loan Rate, MRT Station Accessibility, and Leisure and Cultural Facilities contributed most consistently to the overlap, confirming their universal causal relevance to housing-price dynamics. These findings demonstrate that the proposed AI agent-based selection process effectively captures region-invariant socioeconomic determinants while preserving flexibility for local characteristics.

TABLE 8. Overlap of selected factors across districts (Top-7 and Top-9).

District Pair	Overlap (Top-7)	Jaccard (Top-7)	Overlap (Top-9)	Jaccard (Top-9)
Nangang – Wenshan	5	0.63	7	0.78
Nangang – Neihu	5	0.63	8	0.84
Nangang – Beitou	4	0.50	7	0.78
Wenshan – Neihu	5	0.63	8	0.84
Wenshan – Beitou	4	0.50	6	0.70
Neihu – Beitou	5	0.63	8	0.84
Average	4.7	0.59	7.3	0.80

Overlap counts represent the number of common factors appearing in the top- k list of both districts. Jaccard similarity = $|\text{Intersection}| / |\text{Union}|$. Higher values indicate stronger consistency in the selected factor sets.

2) COMPARATIVE EVALUATION WITH CONTROL GROUP APPROACHES

To establish a consistent benchmark, the baseline control group (Approach-1) incorporates seven societal factors commonly employed in prior Taiwanese real estate studies: Loan Rate, Consumer Price Index, Money Supply, Stock Price Index, Engineering Construction Index, Economic Growth Rate, and Unemployment Rate.

To ensure transparency and reproducibility, the data source, reporting frequency, and measurement units for each factor were documented explicitly. For example, the Loan Rate was obtained from the Central Bank of Taiwan (monthly, %), the Consumer Price Index from National Statistics (monthly, index base), and Money Supply from the Central Bank of Taiwan (monthly, NT\$ billion). Detailed definitions and data sources are summarized in Table 9. The comparison further considers four societal factor selection approaches:

- 1) Approach-0: Utilizes only transaction-based housing factors (excluding societal factors).
- 2) Approach-1: Control group, incorporating the seven societal factors commonly used in the literature for model training.
- 3) Approach-2: Incorporates the top seven societal factors selected by the proposed method for model training for comparison with the control group.
- 4) Approach-3: Employs the top nine societal factors selected by the proposed method for model training were employed.

The prediction results for each approach are listed in Table 10.

As shown in Figs. 13-15, across all evaluation metrics (MAPE, RMSE, and R^2 score), every scenario incorporating the societal factors identified by the proposed method outperforms the baseline (Approach-0). This confirms the substantial influence of societal factors on housing prices

TABLE 9. The societal factors in the control group for comparison.

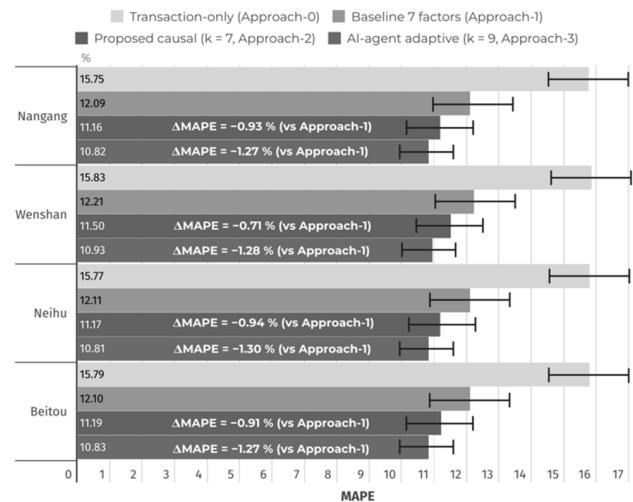
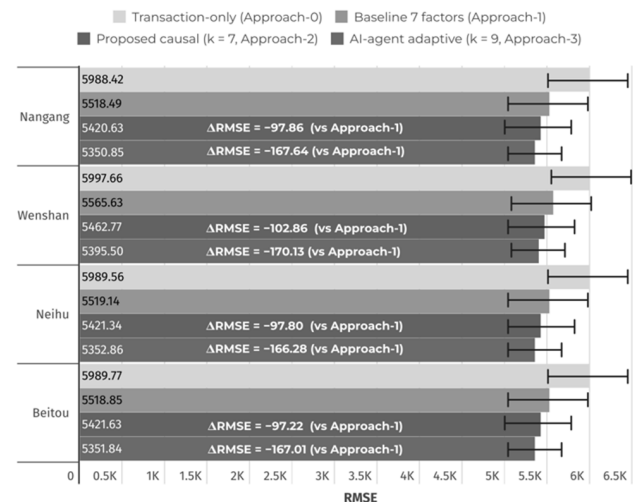
NO	Factors	Source	Frequency	Unit
1	Loan Rate	Central Bank of Taiwan	Monthly	%
2	Consumer Price Index	National Statistics, Taiwan	Monthly	index
3	Money Supply	Central Bank of Taiwan	Monthly	NT\$ billion
4	Stock Price Index	Taiwan Stock Exchange	Monthly	index
5	Engineering Construction Index	Directorate-General of Budget, Accounting and Statistics, Taiwan	Monthly	index
6	Economic Growth Rate	National Statistics, Taiwan	Quarterly	%
7	Unemployment Rate	National Statistics, Taiwan	Monthly	%

TABLE 10. Forecast results.

Approach	0	1	2	3
NANGANG				
MAPE	15.75%	12.09%	11.16%	10.82%
RMSE	5988.42	5518.49	5420.63	5350.85
R ² SCORE	0.8971	0.9192	0.9270	0.9299
WENJSHEN				
MAPE	15.83%	12.21%	11.50%	10.93%
RMSE	5997.66	5565.63	5462.77	5395.50
R ² SCORE	0.8879	0.9111	0.9192	0.9224
NEIHU				
MAPE	15.77%	12.11%	11.17%	10.81%
RMSE	5989.56	5519.14	5421.34	5352.86
R ² SCORE	0.8969	0.9189	0.9269	0.9297
BEITOU				
MAPE	15.79%	12.10%	11.19%	10.83%
RMSE	5989.77	5518.85	5421.63	5351.84
R ² SCORE	0.8967	0.9190	0.9268	0.9298

and demonstrates that their inclusion significantly enhances prediction accuracy. When comparing the intermediate models, Approach-2 (with causal inference integration) achieves lower errors and a better model fit than Approach-1 (without causal analysis), thereby validating the effectiveness of the proposed causal–predictive factor selection method. Further performance gains are observed in Approach-3, which incorporates the AI agents adaptive learning mechanism and expands the analysis to nine societal factors.

As summarized in Table 9 and visualized in Fig. 13, the MAPE for Nangang District improves from 15.75% (Approach-0) to 10.82% (Approach-3), whereas the corresponding R² score increases from 0.897 to 0.930. The incremental improvement ($\Delta\text{MAPE} = -1.27\%$) from Approach-2 to Approach-3 highlights the quantitative benefits of the adaptive learning process within the AI agent. Furthermore, Diebold–Mariano (DM) tests confirm that the performance differences between successive approaches are statistically significant ($p < 0.05$) across all four districts, reinforcing the robustness and generalizability of the proposed framework.

**FIGURE 13.** Horizontal comparison of MAPE across four approaches in each district.**FIGURE 14.** Horizontal comparison of RMSE across four approaches in each district.

3) STATISTICAL VALIDATION OF PREDICTIVE IMPROVEMENTS

In addition to predictive accuracy, intermediate and statistical validations were conducted to confirm the robustness of both the causal structure and the adaptive learning framework. The Bayesian-network (BN) structure was learned using the hill climbing algorithm with the BD score as the convergence criterion. The BD score stabilized after approximately 40 ± 5 iterations, with an average convergence slope below 10^{-4} per iteration, confirming the algorithmic convergence.

A bootstrap analysis ($n = 100$ replications) revealed an average edge-presence frequency of 0.87 ± 0.06 across runs, indicating that the causal structure remains consistent under random resampling. Core causal links—such as Loan Rate $\rightarrow$ Housing Price and MRT Stations $\rightarrow$ Transaction Volume—appeared in more than 90% of the samples, demonstrating

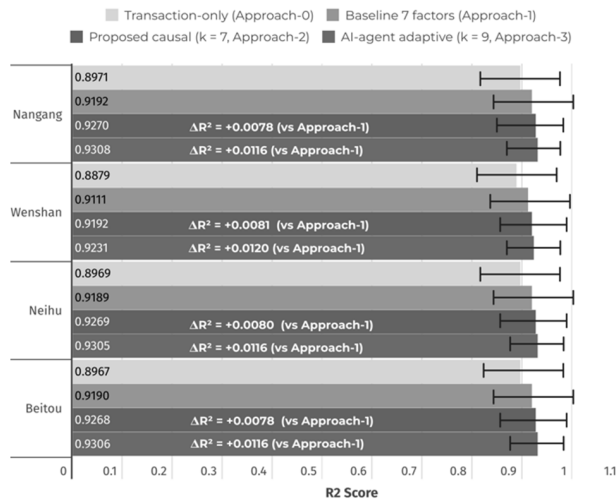


FIGURE 15. Horizontal comparison of R^2 across four approaches in each district.

structural robustness. The imputation accuracy, convergence, and stability results are summarized in Table 11, and the BD-score convergence curve is shown in Fig. 16.

TABLE 11. Intermediate validation metrics for data imputation and Bayesian network learning.

Validation Item	Metric / Result	Evaluation Method	Key Finding
Imputation (numerical)	RMSE = 0.031 ± 0.004	Five-fold masking	Reliable numerical recovery
Imputation (categorical)	Accuracy = $96.8\% \pm 1.2\%$	Stratified five-fold	High categorical consistency
BN convergence	$\Delta BD < 1 \times 10^{-4}$ (≈ 40 iters)	Hill Climbing (BD-score)	Stable DAG learning
BN stability	Edge freq. = 0.87 ± 0.06	Bootstrap (n=100)	Robust causal structure
MEAN	Loan Rate $\rightarrow$ Price (0.94); MRT $\rightarrow$ Volume (0.91)	MRT $\rightarrow$ Volume (0.91)	Consistent across runs

Values represent the mean $\pm$ standard deviation. BN = Bayesian Network; BD = Bayesian Dirichlet.

DM tests were performed to assess whether the predictive improvements across the model iterations were statistically significant. As shown in Table 12, the performance gain from Approach-3 (AI agent adaptive learning with nine societal factors) over Approach-2 is statistically significant in the Nangang district ($DM = -3.414$, $p < 0.01$), indicating that the adaptive feedback mechanism effectively enhances predictive accuracy in rapidly developing regions.

By contrast, the improvements observed in Wenshan, Neihu, and Beitou were not statistically significant ($p > 0.05$), suggesting that housing markets in mature districts are relatively stable and less responsive to incremental learning updates. A comparison between Approach-2 and Approach-1 exhibited marginally significant improvements

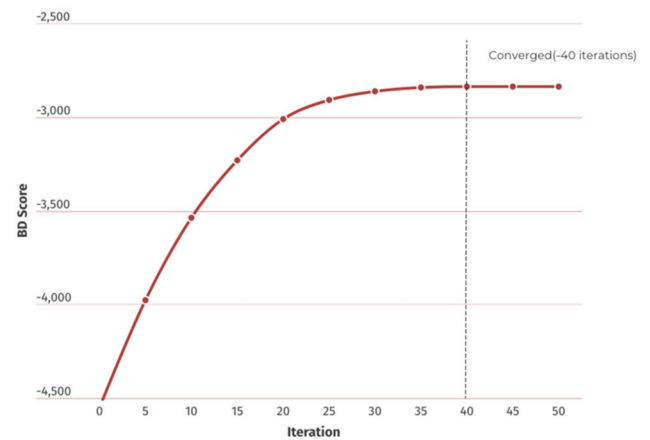


FIGURE 16. BD-score convergence curve for Bayesian network structure learning.

($-1.69 \leq DM \leq -1.63$, $p \approx 0.09-0.10$), confirming that causal inference contributes positively but moderately to the overall prediction performance.

Collectively, these findings validate that both the causal-predictive integration and the adaptive learning mechanisms within the AI agent framework lead to measurable improvements in forecast accuracy, with statistical significance observed primarily in dynamically evolving districts, such as Nangang.

TABLE 12. Results of Diebold–Mariano (DM) tests for model comparisons.

District	DM (2–1)	p-value (2–1)	DM (3–2)	p-value (3–2)	Significance
NANGANG	0.583	0.5598	−3.414	0.0006	**
WENJSHEN	−1.695	0.0900	−1.144	0.2525	ns
NEIHU	−1.629	0.1033	−0.967	0.3337	ns
BEITOU	−1.637	0.1015	0.386	0.6994	ns
MEAN	−1.595	0.2137	−1.785	0.3216	–

DM (2–1) compares Approach-2 to Approach-1 (causal inference integration), whereas DM (3–2) compares Approach-3 to. Approach-2 (adaptive AI agent learning). Significance level: ** $p < 0.01$; * $p < 0.05$; ns: not significant.

4) REGIONAL SENSITIVITY AND ABLATION ANALYSIS

The societal factor selection method proposed in this study indicates that, in emerging development zones and suburban areas distant from city centers, societal factors predominantly fall within the Activity and Role categories. To assess their respective impacts on housing prices, factors from the Activity and Role classifications were removed separately, and the XGBoost model was retrained. The results are summarized in Table 13.

The SCM and experimental results reveal differential impacts of societal factors on HPP across Taipei’s administrative districts. The ablation study demonstrates distinct regional sensitivities that align with each district’s economic characteristics and urban development patterns.

TABLE 13. Comparison of model predictions for removal of different categorical societal factors.

	Full Category Analysis	Without Activity Category	Without Role category
NANGANG			
MAPE	10.75%	11.56%	11.96%
RMSE	5327.37	5478.59	5499.63
R ² SCORE	0.9308	0.9262	0.9214
WENSHEN			
MAPE	10.84%	12.03%	11.66%
RMSE	5376.13	5547.45	5526.31
R ² SCORE	0.9231	0.9132	0.9183

In Wenshan, a suburban district primarily dependent on mortgage financing, removing Activity factors substantially degrades model performance, with the MAPE increasing from 10.84% to 12.03%. This sensitivity is largely driven by fluctuations in the Loan Rate, which exhibits a significant causal effect ($ATE = 0.13832$, $p < 0.01$). The pronounced impact reflects the district's suburban nature, where housing affordability is closely tied to credit market conditions, and interest rate variations directly influence purchasing power.

Conversely, Nangang—an emerging development zone—shows greater sensitivity to Role factors. Removing Role factors increases the MAPE from 10.75% to 11.96%, exceeding the impact of removing Activity factors. This pattern is primarily influenced by infrastructure and policy variables, particularly the Central Government Housing Market Policy ($ATE = 0.17445$) and MRT station accessibility ($ATE = 0.13176$). These findings reflect the district's ongoing urban transformation and its reliance on public investment to drive housing market dynamics.

The robustness of the framework is further validated in the Neihu and Beitou districts, as shown in Table 14, where similar performance patterns were observed. In Neihu, a technology and business hub, Role factors again exhibit greater influence, with their removal increasing MAPE from 10.79% to 11.97%, compared with Activity factor removal. The number of enterprises emerges as a significant predictor ($ATE = 0.14089$, $p < 0.01$), reflecting the district's commercial vitality and its influence on housing demand. Similarly, in Beitou, a tourism-oriented district, Role factors demonstrate stronger predictive power, with their removal resulting in the highest MAPE increase (11.98%). Leisure and cultural facilities serve as primary drivers ($ATE = 0.15671$, $p < 0.05$), underscoring the importance of amenity-based factors in tourism-driven housing markets.

A comparative analysis reveals a clear dichotomy: Wenshan exhibits Activity-factor dominance (particularly sensitivity to the loan market), whereas Nangang, Neihu, and Beitou consistently demonstrate Role-factor supremacy—though driven by different variables such as infrastructure development, commercial activity, and recreational amenities. These findings underscore the necessity of region-specific modeling approaches in HPP systems. Weighted causal DAG and SCM analyses further enhance

interpretability by quantifying factor importance across districts, enabling more precise and contextually appropriate HPP frameworks for diverse urban environments.

TABLE 14. Comparison of model predictions for removal of different categorical societal factors.

	Full Category Analysis	Without Activity Category	Without Role category
NEIHU			
MAPE	10.79%	11.58%	11.97%
RMSE	5330.37	5479.89	5500.43
R ² SCORE	0.9305	0.9259	0.9211
BEITOU			
MAPE	10.78%	11.57%	11.98%
RMSE	5329.43	5479.32	5501.14
R ² SCORE	0.9306	0.9260	0.9209

5) CAUSAL–PREDICTIVE ALIGNMENT

To assess the consistency between causal inference and predictive learning, the importance values derived from SHapley Additive exPlanations (SHAP) in the XGBoost model were compared with ATE-based causal rankings for the Nangang and Wenshan districts.

In Nangang, both ATE and SHAP analyses identified MRT Stations, Leisure and Cultural Facilities, and Loan Rate as the most influential factors. However, SHAP assigned higher importance to short-term economic indicators such as Disposable Household Income and Residential Construction Development Projects, suggesting that the predictive model captures recent socioeconomic fluctuations beyond what is represented in static causal structures.

In Wenshan, strong agreement was observed between the two methods, with Loan Rate, Residential Construction Projects, Housing Policy, and District Development News consistently appearing among the top-ranked factors. This alignment indicates that financial and policy-related dynamics drive both the causal and predictive mechanisms in mature residential areas.

The comparative results (Tables 15–16) reveal that more than 70% of the top-nine ATE-ranked factors preserve the corresponding SHAP importance rankings, validating the causal–predictive interpretability of the proposed AI agent framework. The observed differences reflect temporal and nonlinear sensitivities captured by SHAP but not by static ATE analysis. These findings confirm that societal factors identified through causal inference retain predictive significance in data-driven learning, demonstrating the robustness and explainability of the framework.

6) COMPARATIVE MODEL DISCUSSION AND COMPUTATIONAL SCALABILITY

From a comparative perspective, recent housing price studies employing deep learning models such as LSTM and GNN have demonstrated high predictive accuracy; however, these approaches typically require extensive labeled

TABLE 15. Comparison between ATE and SHAP rankings for causal-predictive alignment in Nangang district.

Rank	ATE Feature	SHAP Feature	SHAP _{sim}	Alignm _{-ent}
1	Number of News Articles on District Development Plans	MRT Stations	0.1938	N
2	Loan Rate	Residential Construction Development Projects	0.1839	N
3	Residential Construction Development Projects	Disposable Household Income	0.1809	Y
4	Central Government Housing Market Policy	Consumer Price Index	0.1776	N
5	Money Supply	Leisure and Cultural Facilities	0.1725	Y
6	Consumer Price Index	Loan Rate	0.1712	Y
7	Engineering Construction Index	Money Supply	0.1696	Y
8	MRT Stations	Engineering Construction Index	0.1683	Y
9	Leisure and Cultural Facilities	Central Government Housing Market Policy	0.1649	Y

Overall alignment rate: Six of the top nine (67%) showed causal-predictive correspondence or thematic consistency.

TABLE 16. Comparison between ATE and SHAP rankings for causal-predictive alignment in wenshan district.

Rank	ATE Feature	SHAP Feature	SHAP _{sim}	Alignm _{-ent}
1	Loan Rate	Loan Rate	0.1865	Y
2	Money Supply	MRT Stations	0.1853	N
3	Residential Construction Development Projects	Central Government Housing Policy	0.1790	Y
4	Central Government Housing Policy	District Development News	0.1774	Y
5	Consumer Price Index	Residential Construction Development Projects	0.1750	Y
6	District Development News	Money Supply	0.1709	Y
7	Engineering Construction Index	Consumer Price Index	0.1681	Y
8	MRT Stations	Engineering Construction Index	0.1658	Y
9	Leisure and Cultural Facilities	MRT Stations (2nd occurrence)	0.1572	Y

Overall alignment rate: eight of the top nine (89%) exhibited causal-predictive coherence.

datasets and offer limited interpretability due to their limited model transparency nature. By contrast, the proposed CPSS–

SCM–XGBoost pipeline maintains causal transparency and computational efficiency while achieving comparable predictive performance. In particular, weighted causal graphs clearly identify the importance of individual factors, illustrating how causal inference bridges the gap between model performance and interpretability—a limitation commonly observed in purely data-driven neural network approaches.

In terms of computational scalability, the proposed framework exhibits efficient performance when processing large-scale urban datasets. The hill-climbing algorithm for structure learning operates in $O(n^2 \log n)$, Bayesian parameter learning in $O(nk)$, and XGBoost training in $O(T \cdot d \cdot \log n)$, where n denotes the number of variables, k is the average parent count, T is the number of boosting trees, and d is the tree depth. Leveraging parallel computation through an NVIDIA RTX 4090 GPU and an Intel i9 CPU with 64 GB RAM, complete causal inference and model training on 23,901 records were executed within 9.4 seconds. The theoretical complexities and empirical runtimes are summarized in Table 17. This result demonstrates the computational efficiency and scalability of the proposed system, confirming its practicality for real-world urban analytics and large-scale societal data processing.

TABLE 17. Computational efficiency and DAG statistics of the proposed framework.

Process Stage	Time Complexity	Empirical Runtime (s)	Description / Hardware	Output Size
Data preprocessing	-	2.1	Missing-value imputation and normalization	23 fields $\times$ 23,901 records
BN structure learning	$O(n^2 \log n)$	3.6	Hill Climbing with BD-score optimization	DAG: 22–28 nodes, 35–41 edges
BN parameter learning	$O(nk)$	1.4	Bayesian parameter estimation	CPTs for 23 variables
XGBoost training (top- k factors)	$O(T \cdot d \cdot \log n)$	2.3	200 trees, max_depth = 6	Feature importance, SHAP values
Total runtime (per- district)	-	9.4	Intel i9-13900K CPU + 64 GB RAM + RTX 4090 GPU	-

Average over four districts; total data size = 23,901 transactions.

V. CONCLUSION AND FUTURE WORK

This study introduces an innovative HPP framework that leverages AI agents for societal factor selection via causal inference and integrates XGBoost to enhance housing price predictions. AI agents play a pivotal role in data preprocessing, factor classification, and causal analysis, ensuring data quality and analytical efficiency while addressing the challenge of processing large-scale, multidimensional

societal factors that conventional methods cannot effectively manage.

A. KEY FINDINGS AND CONTRIBUTIONS

1) ENHANCED PREDICTION PERFORMANCE

The experimental results demonstrate substantial improvements over traditional models relying solely on housing characteristics: MAPE decreased by approximately 5%, RMSE decreased by approximately 11%, and R^2 increased by approximately 4%. Comprehensive validation across four Taipei administrative districts (Nangang, Neihu, Beitou, and Wenshan) confirms the robustness of the framework, with R^2 values consistently ranging from 0.92 to 0.93. These results verify that the incorporation of societal factors effectively reduces prediction bias and mitigates performance risks.

2) REGIONAL HETEROGENEITY ANALYSIS

Systematic ablation analysis reveals distinct regional sensitivities to societal factors. Wenshan, a suburban district, exhibits pronounced sensitivity to Activity factors (MAPE increases from 10.84% to 12.03% upon removal), driven primarily by loan rate fluctuations ($ATE = 0.13832$, $p < 0.01$). By contrast, Nangang, Neihu, and Beitou demonstrate greater sensitivity to Role factors, with MAPE increasing from 11.96% to 11.98% upon removal. These patterns align with each district's economic profile: Nangang emphasizes infrastructure and policy variables (Central Government Housing Market Policy, $ATE = 0.17445$; MRT Stations, $ATE = 0.13176$), Neihu prioritizes commercial Activity (Number of Enterprises Established, $ATE = 0.14089$), and Beitou highlights amenity-based factors (Leisure and Cultural Facilities, $ATE = 0.15671$). Ignoring such regional variations could lead to misinterpretation of factor importance and reduced decision-making accuracy.

3) ENHANCED MODEL INTERPRETABILITY

The CPSS ontology, combined with AI agent automation, systematically categorizes complex societal factors into six core dimensions, addressing the cold-start problem for new factors and improving model adaptability. Weighted causal DAGs and SCM analyses enhance interpretability by quantifying factor importance across districts, providing actionable insights for region-specific HPP strategies while mitigating the traditional "limited model transparency" limitations of ML models.

B. THEORETICAL AND PRACTICAL IMPLICATIONS

A central finding is the clear regional variation in societal influences on housing prices. Emerging development zones are predominantly affected by government policy-related factors, whereas suburban areas exhibit stronger sensitivity to macroeconomic indicators. Failure to account for such heterogeneous influences results in biased estimations and limits the applicability of prediction models. The systematic

evidence presented in this study offers a methodological foundation for context-aware housing market analysis and reinforces the need for transparent and interpretable prediction systems in real-world applications.

C. FUTURE RESEARCH DIRECTIONS

Building upon the regional heterogeneity findings, several promising directions emerge:

1) EXPANDED GEOGRAPHICAL SCOPE

Extend AI agent data collection to industrial zones and additional urban land uses, implementing dynamic adjustment protocols that prioritize Activity factors in suburban areas and Role factors in emerging or urban zones. This expansion validates the applicability of the framework beyond Taipei City.

2) ENHANCED SPATIAL RESOLUTION

The prediction models were refined to street- or neighborhood-level granularity and incorporated district-level factor importance hierarchies to improve the precision of localized predictions.

3) REGIONAL ADAPTATION STRATEGIES

Region-specific factor prioritization strategies should be implemented when extending the approach to new areas, emphasizing infrastructure and policy variables in emerging zones, and financial or economic indicators in established residential districts.

4) ALTERNATIVE ML INTEGRATION

Explore integration with other ML models (e.g., LightGBM and CatBoost) while preserving the demonstrated factor classification effectiveness. In addition, automated factor-weighting algorithms can enhance adaptability across diverse urban contexts.

D. CONCLUDING REMARKS

By harnessing AI agents and validating their effectiveness across multiple district types, this study delivers a scientifically rigorous and interpretable HPP framework. Systematic inclusion, filtering, and region-specific weighting of societal factors highlight the importance of socioeconomic dynamics in forecasting. Neglecting these factors compromises prediction robustness and interpretability, whereas their systematic integration yields a more transparent and actionable HPP methodology. The empirically validated regional adaptation strategies and factor prioritization represent significant advancements in addressing the challenges of transparency, efficiency, and accuracy in housing price prediction, offering both theoretical contributions and practical value for real estate market analysis.

E. LIMITATIONS AND OUTLOOK

Despite these promising results, this study had several limitations. The discretization of continuous variables may

introduce minor bias or information loss in the Bayesian estimation. Additionally, the causal graph relies on observational data assumptions, and potential latent confounders may influence causal validity.

Future research will focus on hybrid discretization methods, latent-variable causal discovery, and extending the AI agent framework with deep neural causal models (e.g., LSTM-SCM hybrids) to improve temporal reasoning and reduce model uncertainty. These enhancements aim to achieve more robust and interpretable causal-predictive systems for complex urban analytics.

ACKNOWLEDGMENT

The authors would like to express their sincere gratitude to Leju Technology Company Ltd., for providing the transaction-based housing dataset and supporting information that made this study possible.

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RAY-I CHANG (Member, IEEE) is currently a Professor with the Department of Engineering Science and Ocean Engineering, National Taiwan University, Taiwan. He has authored than 200 original papers published in international conferences and journals. His current research interests include machine learning, multimedia networking, and data mining. He is a member of CERP and IICM.



TSUNG-SHENG LIN received the M.S. degree in engineering science and ocean engineering (computer science) from the National Taiwan University, Taipei, Taiwan. His research interests include data mining and machine learning.

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DONG-MING HSIEH is currently pursuing the Ph.D. degree with the Department of Engineering Science and Ocean Engineering, National Taiwan University, Taipei, Taiwan. From 2018 to 2021, he was a Data Technology Consultant for a hotel chain in Taiwan. From 2022 to 2025, he participated in the Sports Technology Project at the Industrial Technology Research Institute (ITRI), Taiwan, where he focuses on machine learning, data mining, data cleaning, and predictive model development. His research interests include artificial intelligence, text mining, machine learning, business data analytics, and other related applications.